

Developmental and Epileptic Encephalopathy 80 (DEE80 / PIGB-related GPI Deficiency): A Comprehensive Disease Report

Summary

Developmental and Epileptic Encephalopathy 80 (DEE80; OMIM #618580; also EIEE80, GPI biosynthesis defect 21) is an ultra-rare, autosomal-recessive **inherited glycosylphosphatidylinositol (GPI) deficiency disorder (IGD)** caused by biallelic loss-of-function variants in **PIGB** (phosphatidylinositol glycan anchor biosynthesis class B; HGNC:8959; chromosome 15q21.3). PIGB encodes the endoplasmic-reticulum mannosyltransferase that transfers the **third mannose** to the growing GPI core glycan. Because roughly 150 human cell-surface proteins are attached to the plasma membrane through a GPI anchor — many of them adhesion, guidance and signaling molecules essential for brain development — the loss of PIGB function reduces surface presentation of all of them simultaneously. The clinical consequence is a severe neurodevelopmental disorder combining **early-onset, often intractable epilepsy, global developmental delay/intellectual disability, hypotonia, axonal peripheral neuropathy**, variable **brain malformations** (polymicrogyria, hypomyelination) and, in most patients, **elevated serum alkaline phosphatase (hyperphosphatasia)**, with high mortality in early childhood in severe cases.

The disease was first delineated by Murakami et al. in 2019, who described 10 unrelated families with biallelic PIGB mutations; all affected individuals had seizures, most had developmental/intellectual delay, and eight children died before four years of age. Subsequent case reports (e.g., Schiavoni 2021) and large cross-gene IGD cohorts (Bellai-Dussault 2019; Sidpra 2024) have broadened the phenotypic spectrum from lethal neonatal encephalopathy to milder presentations with survival into later childhood, and have defined a shared neuroimaging signature including cerebral/cerebellar atrophy and symmetric restricted diffusion of the **central tegmental tracts**.

This report synthesizes eight confirmed findings and 29 reviewed papers into a coherent mechanistic and clinical picture. The two most mechanistically satisfying insights are: (1) **why serum alkaline phosphatase is elevated** — PIGB deficiency accumulates an *incomplete but mannose-bearing* GPI that the transamidase still attaches to, resulting in secretion (shedding) rather than membrane anchoring of tissue-nonspecific alkaline phosphatase (TNAP); and (2) **why vitamin B6 partially helps seizures** — TNAP is the ectoenzyme that dephosphorylates circulating pyridoxal-5'-phosphate (PLP), so perturbed surface TNAP disturbs vitamin B6 handling, providing a biochemical rationale for supraphysiologic pyridoxine/PLP supplementation. There is no cure; management is supportive with mechanism-based, partially effective high-dose pyridoxine/pyridoxal-5'-phosphate for seizures.

1. Disease Information

Overview. DEE80 is a Mendelian, autosomal-recessive congenital disorder of glycosylation belonging to the inherited GPI deficiency (IGD) subgroup. It is a developmental and epileptic encephalopathy — meaning both the underlying developmental disturbance and the recurrent epileptic activity contribute to the encephalopathy — of prenatal/early-infantile onset.

Key identifiers:

Resource	Identifier
OMIM (phenotype)	#618580 (Developmental and epileptic encephalopathy 80)
Gene	PIGB — OMIM *604122; HGNC:8959; NCBI Gene 9488; Ensembl ENSG00000069943; UniProt Q92521
Locus	15q21.3
Suggested MONDO	Inherited GPI deficiency / DEE80 (MONDO term for EIEE80)
ICD-11	8A61 (Developmental and epileptic encephalopathies) — closest applicable code
Inheritance	Autosomal recessive

Synonyms / alternative names: EIEE80 (Epileptic encephalopathy, early infantile, 80); Developmental and epileptic encephalopathy 80; PIGB-related GPI biosynthesis defect; GPI biosynthesis defect 21 (GPIBD21); PIGB-CDG. Note that a broader allelic spectrum exists: biallelic *PIGB* variants have also been linked to **Acrofrontofacionasal dysostosis type 1 (AFFND1)** (PMID: 34400385), and a distinct constitutional *PIGB* mechanism can predispose to **paroxysmal nocturnal hemoglobinuria (PNH)** via somatic copy-number-neutral loss of heterozygosity (PMID: 33216889).

Source of information. Disease-level aggregated resources (OMIM, Orphanet) combined with individual-patient primary literature (case series and cohort studies). The founding evidence is individual-patient data from ~16 affected individuals across 10 families (PMID: 31256876), augmented by cross-gene IGD cohorts of 202 (PMID: 30054924) and 83 individuals (PMID: 38456468).

2. Etiology

Primary cause — genetic (Finding F001). DEE80 is caused exclusively by **biallelic (homozygous or compound heterozygous) pathogenic variants in PIGB**. Murakami et al. (2019) described ten unrelated families each carrying different PIGB mutations (10 distinct variants), and demonstrated by flow cytometry that blood cells and fibroblasts had **decreased cell-surface GPI-anchored proteins**, confirming causality.

"We describe ten unrelated families with bi-allelic mutations in PIGB, a gene that encodes phosphatidylinositol glycan class B, which transfers the third mannose to the GPI." — PMID: 31256876

"Flow cytometric analysis of blood cells and fibroblasts from the affected individuals showed decreased cell surface presence of GPI-anchored proteins." — PMID: 31256876

Genetic risk factors. The only established risk factor is inheritance of two damaging PIGB alleles. Consanguinity increases risk of homozygosity (several reported families are consanguineous). No common susceptibility loci or GWAS signals apply — this is a monogenic disorder.

Environmental / infectious factors. No environmental, toxic, infectious, or lifestyle cause contributes to disease onset. As with other IGDs, **fever and intercurrent infection can worsen seizures** and precipitate status epilepticus (documented in the related PIGW disorder; [PMID: 38055078](#)), but these are triggers of symptoms, not causes of disease.

Protective factors. None genetically defined. Supraphysiologic vitamin B6 (pyridoxine/pyridoxal-5'-phosphate) mitigates seizures in a subset (see Treatment).

Gene–environment interactions. Limited; the principal clinically relevant interaction is fever/infection as a seizure-provoking factor superimposed on the genetic lesion.

3. Phenotypes

DEE80 phenotypes span neurological, metabolic and multisystem domains. Frequencies below combine the founding PIGB cohort ([PMID: 31256876](#)), a milder single case ([PMID: 34161862](#)), and cross-gene IGD cohorts ([PMID: 30054924](#); [PMID: 38456468](#)).

Phenotype	Type	Onset	Frequency	Suggested HPO
Seizures / epilepsy	Clinical sign	Early infantile (median ~6 mo in IGDs)	~100% (PIGB cohort); 83% (IGD cohort)	HP:0001250 (Seizure); HP:0011097 (Epileptic spasms)
Global developmental delay / intellectual disability	Clinical sign	Congenital/ infantile	Most; 90% (IGD cohort)	HP:0001263; HP:0001249
Hypotonia	Clinical sign	Infantile	72% (IGD cohort)	HP:0001252
Motor symptoms	Clinical sign	Infantile	64% (IGD cohort)	HP:0100022
Peripheral (axonal) neuropathy	Clinical sign	Childhood	4/~16 (PIGB)	HP:0000762; HP:0003477
Polymicrogyria	Physical/ imaging	Congenital	2/~16 (PIGB)	HP:0002126
Hypomyelination	Imaging	Congenital/ infantile	Reported (milder case)	HP:0006808
Elevated alkaline phosphatase (hyperphosphatasia)	Laboratory	Congenital	Most	HP:0003155
2-oxoglutaric aciduria	Laboratory	—	2 severe cases (PIGB)	HP:0003355 (Organic aciduria)
Scoliosis	Physical	Childhood	Reported (milder case)	HP:0002650
Foot deformity (equino-varo-cavus)	Physical	Childhood	Reported (milder case)	HP:0001760
Dysmorphic features	Physical	Congenital	82% (IGD cohort), variable	HP:0001999

Phenotype	Type	Onset	Frequency	Suggested HPO
Early mortality	Outcome	<4 years	8 children (PIGB); 15/83 (IGD)	—

Phenotype characteristics (Finding F002). In the founding cohort, "most of the affected individuals have global developmental and/or intellectual delay, all had seizures, two had polymicrogyria, and four had a peripheral neuropathy. Eight children passed away before four years old" (PMID: 31256876). Two severely affected individuals showed **2-oxoglutaric aciduria**, indicating a metabolic derangement in the most severe end of the spectrum. Two individuals carried a clinical diagnosis of **DOORS syndrome** (Deafness, Onycho-Osteodystrophy, mental Retardation, Seizures) before the molecular cause was known.

Severity is **variable**. Schiavoni et al. (2021) reported the milder end: "severe global developmental delay with absent speech, mixed peripheral polyneuropathy, hypotonia, bilateral equino-varo-supinated-cavus foot, early-onset scoliosis, elevated serum alkaline phosphatase and a single episode of febrile status epilepticus. Hypomyelination was documented on brain MRI" (PMID: 34161862). Progression is generally **static-to-progressive**; Murakami & Kinoshita note the disease "progresses even after birth" (PMID: 26165085).

Quality-of-life impact. Profound. Affected children typically have absent or minimal speech, are non-ambulatory or motor-impaired, require anticonvulsant therapy, and depend on full caregiving. No disease-specific EQ-5D/SF-36 data exist given rarity.

4. Genetic / Molecular Information

Causal gene. *PIGB* — phosphatidylinositol glycan anchor biosynthesis, class B (HGNC:8959; NCBI Gene 9488; OMIM 604122; UniProt Q92521; 15q21.3). *PIGB* is a *GT-C superfamily mannosyltransferase** (GPI mannosyltransferase III / GPI-MT-III) that transfers the third mannose (Man3) onto the GPI intermediate in the ER membrane (PMID: 31256876; topology reviewed in PMID: 31569500). It shares a conserved membrane-embedded "BindGPILA" domain with PIG-M, PIG-V, PIG-Z, PIG-U and PIG-W (PMID: 29764287).

Pathogenic variants. At least 10 distinct biallelic variants were reported in the founding families, including missense and predicted loss-of-function alleles; an intronic splice variant causing exon skipping (a null allele) was found in AFFND1 families (PMID: 34400385). Variant

classification follows ACMG/AMP; most are classified pathogenic/likely pathogenic in ClinVar. Functional consequence is **loss of function** — reduced or absent third-mannose transfer, leading to reduced GPI-anchored protein surface expression ([PMID: 31256876](#)). Transfection rescue assays confirm variant PIGB fails to restore GPI-anchored protein surface expression in PIGB-deficient CHO cells ([PMID: 33216889](#)).

Allele frequency. Pathogenic alleles are individually ultra-rare in gnomAD (consistent with a recessive, severe disorder). **Origin:** germline (constitutional). Notably, a constitutional heterozygous PIGB variant rendered homozygous by somatic copy-number-neutral loss of heterozygosity can produce a PNH clone ([PMID: 33216889](#)) — a distinct, non-DEE80 phenomenon.

Modifier genes / epigenetics. No specific modifier genes or epigenetic mechanisms are established for DEE80. Phenotypic variability (lethal vs milder) likely reflects residual PIGB enzymatic activity (hypomorphic vs null alleles), analogous to other IGDs.

Chromosomal abnormalities. Not a feature of DEE80 itself; the associated 70-kbp 15q microdeletion in the PNH report involves TM2D3/TARSL2, unrelated to the encephalopathy ([PMID: 33216889](#)).

Subcellular localization note. PIGB is normally an ER/nuclear-envelope membrane enzyme; work in fungal systems shows nuclear-envelope localization can be essential for its activity ([PMID: 30266758](#); [PMID: 32051283](#)).

5. Environmental Information

DEE80 is a purely genetic disorder. **No environmental toxins, radiation, pollution, occupational exposures, lifestyle factors, or infectious agents cause it.** The only clinically relevant environmental modifiers are **fever and infection**, which can lower seizure threshold and precipitate status epilepticus. Dietary vitamin B6 status is relevant only as a therapeutic lever (see Treatment).

6. Mechanism / Pathophysiology

Ordered causal chain

1. **Biallelic loss-of-function PIGB variants → loss of GPI mannosyltransferase-III activity** in the ER (fails to add the third mannose to the GPI glycan core). (*Demonstrated — enzyme function and rescue assays.*)
2. Loss of Man3 addition → **accumulation of an incomplete, mannose-bearing GPI intermediate** and **globally reduced surface presentation of GPI-anchored proteins (GPI-APs)** on all cell types (shown by flow cytometry). (*Demonstrated.*)
 - 3a. **Branch A (neurodevelopment):** Reduced surface GPI-APs → loss of GPI-anchored adhesion/guidance/signaling molecules (glypicans, contactins, RGMa, and ~150 others) → **impaired cortical progenitor proliferation, neuronal migration, axon guidance and myelination** → polymicrogyria, hypomyelination, axonal neuropathy, developmental delay. (*Mechanistically inferred from the established roles of individual GPI-APs.*)
 - 3b. **Branch B (network excitability):** Developmental disorganization + altered GPI-anchored receptor signaling → **cortical hyperexcitability** → **early-onset, often intractable seizures** → epileptic encephalopathy compounding the developmental deficit. (*Inferred.*)
 - 3c. **Branch C (alkaline phosphatase / vitamin B6):** Because the accumulated GPI still bears mannose, the transamidase attaches TNAP to it, but the incomplete anchor causes **TNAP to be secreted/shed rather than membrane-retained** → **serum hyperphosphatasia** and **reduced surface TNAP** → **perturbed dephosphorylation of pyridoxal-5'-phosphate (PLP)** → disturbed vitamin B6 homeostasis contributing to seizures. (*Branch C "secretion" step demonstrated in CHO cells; the PLP-handling link is biochemically inferred.*)
3. Net result: **developmental and epileptic encephalopathy with multisystem involvement and high early mortality.**

Molecular detail

Loss of GPI-anchored guidance/adhesion proteins drives cortical maldevelopment (Finding F005). Many GPI-APs are master regulators of corticogenesis. Glypicans: **GPC4** (expressed in cortical progenitors) "promotes their proliferation and the generation of intermediate progenitors, whereas neuronal GPC2 acts as a brake on radial neuronal migration" ([PMID: 42275470](#)); the GPI-anchored morphogen receptor **GPC3** forms a complex with Unc5 to "guide migrating pyramidal neurons in the mouse cortex" ([PMID: 36240740](#)). Contactins (**CNTN2/TAG-1**, **CNTN4**) mark and modulate migrating neurons and neurite

elongation (PMID: 40580015; PMID: 38745463; PMID: 30629639), and **RGMa** is a GPI-anchored guidance molecule regulating neuronal differentiation and survival (PMID: 36089003). Because PIGB loss reduces surface presentation of *all* GPI-APs simultaneously (PMID: 31256876), these pathways are impaired in parallel — explaining polymicrogyria, migration defects, hypomyelination and neuropathy.

Hyperphosphatasia mechanism (Finding F006). Murakami et al. (2012) directly demonstrated the molecular basis in CHO mutants: "The GPI-anchored protein was secreted substantially into medium from PIGV-, PIGB-, and PIGF-deficient CHO cells, in which incomplete GPI bearing mannose was accumulated. In contrast, ALP was degraded in PIGL-, DPM2-, or PIGX-deficient CHO cells, in which incomplete shorter GPIs that lacked mannose were accumulated" (PMID: 22228761). Thus PIGB deficiency accumulates a *mannose-bearing* (but incomplete) GPI that the transamidase still uses, releasing TNAP into serum — the direct molecular cause of hyperphosphatasia in DEE80.

Vitamin B6 rationale (Finding F007). TNAP is itself a GPI-anchored ectoenzyme that dephosphorylates **pyridoxal-5'-phosphate (PLP)** — the predominant circulating form of vitamin B6 — to pyridoxal, which alone can cross the neuronal membrane to be re-phosphorylated intracellularly. Buchet, Millán & Magne (2013): TNAP deficiency (hypophosphatasia) "leads to ... epileptic seizures in the most severe cases, caused by abnormal metabolism of pyridoxal-5'-phosphate (the predominant form of vitamin B6)" (PMID: 23860646); TNAP's broad substrate range including PLP is confirmed by Imam et al. (2024) (PMID: 39728440). In GPI deficiencies the surface TNAP pool is reduced/shed, perturbing B6 handling — providing the biochemical rationale for supraphysiologic pyridoxine/PLP supplementation, which partially reduces seizures (PMID: 35080266).

Upstream vs downstream, cell types, GO/CL terms

- **Upstream (initiating):** ER GPI biosynthesis defect (GO:0006506 GPI anchor biosynthetic process; GO:0000030 mannosyltransferase activity). Compartment: endoplasmic reticulum (GO:0005783); nuclear envelope (GO:0005635).
- **Downstream (effector):** neuronal migration (GO:0001764), axon guidance (GO:0007411), central nervous system myelination (GO:0022010), regulation of neuron differentiation (GO:0045664); dephosphorylation of PLP (GO:0016791 phosphatase activity).
- **Cell types (CL):** neural progenitor/radial glia (CL:0000031 neuroblast; CL:0000681 radial glial cell), pyramidal neuron (CL:0000598), oligodendrocyte (CL:0000128), peripheral neuron/Schwann-associated axons (CL:0002573).

7. Anatomical Structures Affected

Organ / system level. Primary organ: **brain** (central nervous system) — cerebral cortex, white matter, cerebellum, brainstem. Secondary: **peripheral nervous system** (axonal neuropathy). Multisystem involvement is common in IGDs — Sidpra et al. found systemic involvement in 61/83, with gastrointestinal 66%, cardiac 19%, renal 14% ([PMID: 38456468](#)). Musculoskeletal features (scoliosis, foot deformity) reflect neuromuscular and connective-tissue involvement.

Neuroimaging signature (Finding F008). In the largest IGD cohort, prognostically significant features were "cerebral atrophy (75%), cerebellar atrophy (60%), callosal anomalies (57%) and **symmetric restricted diffusion of the central tegmental tracts (60%)**" ([PMID: 38456468](#)). Polymicrogyria and hypomyelination are specifically documented in PIGB patients.

Tissue / cell level. Nervous tissue predominantly: cortical neurons, radial glia/progenitors, oligodendrocytes (hypomyelination), peripheral axons. **Subcellular:** the initiating defect is in the **endoplasmic reticulum** (GPI assembly) and **plasma membrane** (GPI-AP presentation).

UBERON terms: UBERON:0000955 (brain); UBERON:0000956 (cerebral cortex); UBERON:0002037 (cerebellum); UBERON:0002316 (white matter); UBERON:0002298 (brainstem tegmentum, central tegmental tract region); UBERON:0000044 (peripheral nerve / dorsal root ganglion). **Lateralization:** bilateral/symmetric (e.g., symmetric central tegmental tract diffusion restriction; bilateral foot deformity).

8. Temporal Development

Onset. Congenital to early-infantile. In IGDs the **median age at seizure onset is 6 months** ([PMID: 38456468](#)); developmental delay is apparent from infancy. Onset pattern is chronic/insidious for developmental features and can be acute for seizures.

Progression. The disorder is **progressive after birth** — "the disease progresses even after birth" ([PMID: 26165085](#)) — arguing for early diagnosis and treatment. Course ranges from rapidly fatal (8 children dead before age 4 in the founding cohort; 15/83 deceased in the IGD cohort) to a more static but severely disabled trajectory in milder cases.

Patterns. Seizures are often intractable; no spontaneous remission is described. **Critical periods:** the prenatal/early-infantile window of corticogenesis and myelination is when the irreversible structural damage occurs, and the early-infantile period is the therapeutic window for seizure control (pyridoxine). Fever/infection are episodic aggravators.

9. Inheritance and Population

Epidemiology. Ultra-rare; exact prevalence/incidence are not established. Fewer than ~20 PIGB-specific patients are reported in the literature; DEE80 sits within the broader IGD group (>200 individuals reported across all GPI-pathway genes). No population-level prevalence figure exists.

Inheritance. Autosomal recessive. Requires two pathogenic PIGB alleles. **Penetrance** appears complete for biallelic loss-of-function. **Expressivity is variable** (lethal neonatal to milder childhood phenotypes), likely reflecting residual enzyme activity. **Genetic anticipation:** not applicable (no repeat expansion). **Germline mosaicism:** not specifically reported. **Consanguinity** contributes (homozygous cases in consanguineous families). **Carrier frequency:** not established; individually rare alleles.

Population demographics. Reported families are geographically and ethnically diverse (no established founder population for PIGB). **Sex ratio:** autosomal — no sex bias expected. **Age distribution:** affected individuals are infants and young children; survival beyond childhood occurs in milder cases.

10. Diagnostics

Biochemical clues (Finding F003). Elevated serum alkaline phosphatase (**hyperphosphatasia**) is a hallmark and a strong pointer to IGD: "The presence of hyperphosphatasia is strong evidence of IGD. Flow cytometric analysis of GPI-APs on granulocytes is also useful for the detection of IGD" (PMID: 25803904). Clinical-clue combinations — "Certain combinations, such as seizures with aplastic/hypoplastic nails or abnormal alkaline phosphatase levels suggest an inherited GPI deficiency" (PMID: 30054924). Note: not all IGDs show hyperphosphatasia; some show normal/low ALP depending on where the biosynthetic block lies.

Flow cytometry. Reduced surface GPI-anchored proteins (e.g., CD16, CD24, FLAER, CD59) on granulocytes/blood cells is a functional confirmatory assay (PMID: 31256876; PMID: 25803904).

Genetic testing (definitive). Diagnosis is confirmed by **whole-exome (WES) or whole-genome (WGS) sequencing** identifying biallelic PIGB variants; WGS/RNA analysis is valuable for detecting deep-intronic/splice variants (e.g., the exon-skipping intronic allele in AFFND1,

[PMID: 34400385](#)). Targeted gene panels for epileptic encephalopathy / congenital disorders of glycosylation that include the PIG genes are appropriate first-line. Segregation studies confirm biallelic inheritance.

Imaging. Brain MRI may show polymicrogyria, hypomyelination, cerebral/cerebellar atrophy, callosal anomalies, and **symmetric restricted diffusion of the central tegmental tracts** ([PMID: 38456468](#)).

Electrophysiology. EEG documents epileptiform activity; nerve conduction studies confirm axonal peripheral neuropathy.

Metabolic. Urine organic acids may reveal **2-oxoglutaric aciduria** in severe cases ([PMID: 31256876](#)).

Differential diagnosis. Other IGDs (PIGA, PIGV, PIGN, PIGO, PIGT, PIGS, PIGW, PIGG, etc.), DOORS syndrome, hyperphosphatasia-mental retardation syndrome (Mabry syndrome, PIGV/PIGO), pyridoxine-dependent epilepsy (ALDH7A1), and other early-infantile epileptic encephalopathies. Two DEE80 patients were initially diagnosed clinically as DOORS syndrome.

11. Outcome / Prognosis

Survival / mortality. Guarded, particularly in severe forms: **eight of ~16 affected children in the founding cohort died before age four** ([PMID: 31256876](#)); **15 of 83** individuals in the broad IGD cohort were deceased at reporting ([PMID: 38456468](#)). Milder patients survive into later childhood.

Morbidity / function. Severe lifelong disability — intellectual disability, absent/minimal speech, motor impairment, epilepsy, neuropathy, orthopedic complications (scoliosis, foot deformity). Full caregiver dependence is typical.

Prognostic factors. Severity of the underlying allele (null vs hypomorphic), presence of cortical malformation, seizure intractability, and neuroimaging burden (cerebral/cerebellar atrophy and central tegmental tract diffusion restriction are described as prognostically significant, [PMID: 38456468](#)). 2-oxoglutaric aciduria marks the most severe metabolic end.

12. Treatment

There is **no cure**; management is symptomatic and supportive, with one mechanism-based pharmacologic lever.

Vitamin B6 (pyridoxine / pyridoxal-5'-phosphate) — mechanism-based, partial (Finding F004). In a cohort of 7 GPI-deficiency patients treated with high-dose pyridoxine (20–30 mg/kg/day) or pyridoxal-5'-phosphate: "We observed more than 50% seizure frequency reduction in 2 out of 7 and less than 50% reduction in another 3 out of 7 participants. No participants reached seizure freedom" (PMID: 35080266); no significant EEG change was seen in 6/7. Murakami & Kinoshita emphasize early treatment: "Early diagnosis and treatment are desirable because the disease progresses even after birth and vitamin B6(pyridoxine) is very effective for some patients with intractable seizures" (PMID: 26165085). The biochemical rationale is the TNAP–PLP link (Findings F006, F007). NCIT: Pyridoxine (C793); Pyridoxal Phosphate (C88250).

Antiseizure medications. Standard antiepileptic drugs are used for seizure control, though seizures are frequently intractable. Choice is empirical/supportive; no genotype-specific ASM is validated.

Supportive / rehabilitative care. Physical, occupational and speech therapy; nutritional support; orthopedic management of scoliosis/foot deformity; management of intercurrent infections that provoke seizures; multidisciplinary developmental support. NCIT: Supportive Care (C15277); Physical Therapy (C15359).

Advanced / experimental therapeutics. No approved gene, cell, or RNA therapy exists for DEE80. Gene replacement is a conceptually attractive future avenue (recessive loss-of-function of a single enzyme), but none is in clinical trials for PIGB. **Pharmacogenomics:** not established.

13. Prevention

Primary prevention. Not possible for a spontaneously arising recessive disorder except through reproductive genetic counseling. **Genetic counseling** for at-risk families (25% recurrence risk for carrier couples) is central. **Carrier screening** and, where an affected proband's variants are known, **prenatal diagnosis or preimplantation genetic testing (PGT-M)** can prevent recurrence.

Secondary prevention. Early molecular diagnosis (rapid WES/WGS in neonates with epileptic encephalopathy plus hyperphosphatasia) enables early initiation of pyridoxine and multidisciplinary support during the progressive early period. There is no population newborn-screening test for PIGB deficiency, though hyperphosphatasia on routine chemistry can be an incidental flag.

Tertiary prevention. Prevention of complications — aspiration, infection-triggered status epilepticus, orthopedic deterioration — through proactive management.

14. Other Species / Natural Disease

PIGB is evolutionarily conserved; orthologs and GPI-pathway homologs exist across eukaryotes (mammals, fungi, protozoa). Functional studies of PIG-B localization/activity have been performed in fungal systems (PMID: 30266758; PMID: 32051283), and the shared BindGPILA membrane domain is conserved across PIG-B/M/V/W/U/Z (PMID: 29764287). **Orthologs:** mouse *Pigb* (NCBI Gene 55981). No naturally occurring PIGB-deficiency disease in companion animals or wildlife is catalogued (no OMIA entry identified). GPI biosynthesis is essential and broadly conserved, so complete loss is generally embryonic-lethal in model organisms. No zoonotic relevance.

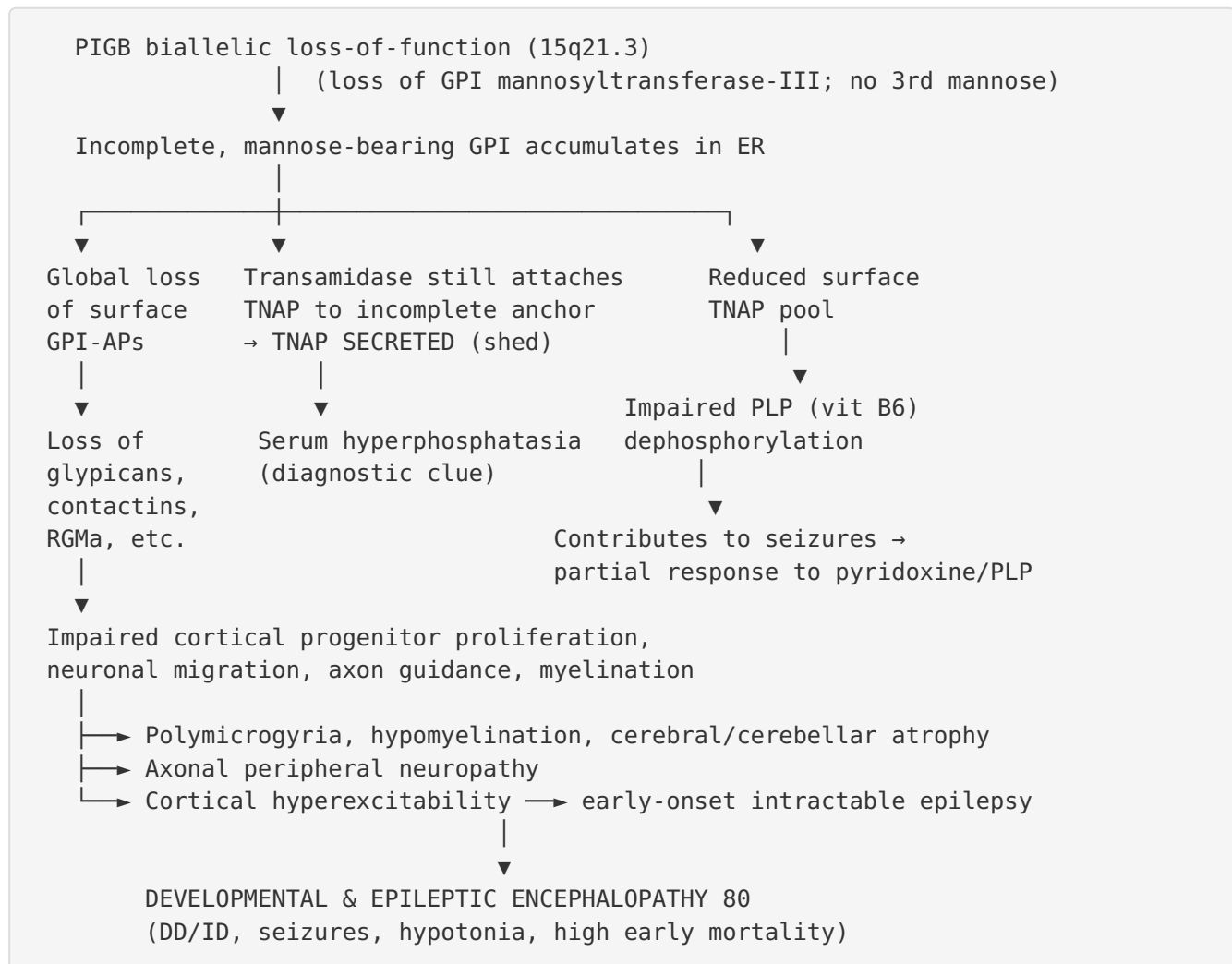
15. Model Organisms

Cellular models (principal experimental system). PIGB-deficient Chinese hamster ovary (CHO) cells are the workhorse for GPI biology and were pivotal in establishing DEE80 mechanism: they show loss of surface GPI-anchored proteins and the diagnostic **secretion of ALP** (PMID: 22228761); transfection rescue confirms variant pathogenicity (PMID: 33216889). PIGB-knockout human cell lines were used to show altered EthN-P bridge usage on GPI anchors (PMID: 35603428). Patient fibroblasts and blood cells serve as primary-cell models for flow-cytometric GPI-AP quantitation (PMID: 31256876).

Genetic model types. Knockout/knock-down cell lines are the primary genetic models. A dedicated mouse model of PIGB deficiency recapitulating DEE80 is not prominently reported (complete GPI-pathway knockouts are typically embryonic lethal; conditional/hypomorphic strategies would be required). MGI lists *Pigb* (mouse), enabling future targeted or conditional models.

Phenotype recapitulation & limitations. Cellular models faithfully reproduce the *biochemical* phenotype (reduced surface GPI-APs; ALP shedding) and are excellent for variant functional testing, but **cannot recapitulate the neurodevelopmental phenotype** (cortical migration, epilepsy, neuropathy). Patient-derived iPSC neurons/organoids would be the logical next step to model corticogenesis defects.

Mechanistic Model / Interpretation



The unifying concept is that **one enzymatic lesion removes an entire class of ~150 cell-surface proteins**, producing a pleiotropic, multi-branch phenotype. Two branches are biochemically demonstrated (GPI-AP loss by flow cytometry; TNAP shedding in CHO cells); the neurodevelopmental branch is inferred from the well-established individual functions of GPI-anchored guidance/adhesion molecules; and the vitamin-B6 branch is a biochemically grounded therapeutic hypothesis supported by partial clinical efficacy.

Evidence Base

PMID	Title (abbrev.)	Evidence type	Role
31256876	<i>Mutations in PIGB Cause an Inherited GPI Biosynthesis Defect...</i>	Human clinical + in vitro	Founding paper ; defines gene, phenotype, mortality, flow-cytometry defect (F001, F002)
34161862	<i>Further delineation of PIGB-related early infantile EE</i>	Human case	Milder-end phenotype (F002)
22228761	<i>Mechanism for release of alkaline phosphatase in GPI deficiency</i>	In vitro (CHO)	Explains hyperphosphatasia (F006)
23860646	<i>Multisystemic functions of alkaline phosphatases</i>	Review	TNAP–PLP–seizure link (F007)
39728440	<i>Structural/functional integration of TNAP</i>	Review	Confirms TNAP dephosphorylates PLP (F007)
35080266	<i>Pyridoxine or PLP for seizures in GPI deficiency</i>	Human cohort (n=7)	Partial efficacy of B6 (F004)
26165085	<i>[Inherited GPI deficiency...]</i>	Review	B6 effective in some; progressive disease (F004)
25803904	<i>[Inherited GPI deficiencies...]</i>	Review	Hyperphosphatasia + flow cytometry diagnostics (F003)
30054924	<i>Clinical variability in IGDs</i>	Human cohort (n=202)	Diagnostic clues (F003)
38456468	<i>Clinical and genetic spectrum of IGDs</i>	Human cohort (n=83)	Natural history, imaging signature (F008)
42275470	<i>Glypican core proteins in corticogenesis</i>	Model/in vitro	GPC4/GPC2 in migration (F005)
36240740	<i>GPC3-Unc5 receptor complex in migration</i>	Model/structural	GPC3 guides cortical neurons (F005)

Model

PMID	Title (abbrev.)	Evidence type	Role
40580015 , 38745463 , 30629639	Contactins CNTN2/ CNTN4/CNTN1		Migrating-neuron / neurite roles (F005)
36089003	<i>RGMa...</i>	Model	GPI-anchored guidance/survival molecule (F005)
34400385	<i>Intronic PIGB variant in AFFND1</i>	Human + in vitro	Allelic spectrum; null via exon skipping
33216889	<i>PNH from CN-LOH of constitutional PIGB</i>	Human + in vitro	Rescue assay; distinct PNH mechanism
29764287 , 31569500 , 30266758 , 32051283	PIG-B structure/ topology/localization	Computational/ model	Enzyme biology context
35603428	<i>EthN-P on second mannose...</i>	In vitro	PIGB-KO GPI biology
30269814 , 32198969 , 38055078	PIGS/PIGW IGDs	Human	Related IGDs; differential diagnosis, B6-responsive seizures

Limitations and Knowledge Gaps

- 1. Small n.** Fewer than ~20 molecularly confirmed PIGB/DEE80 patients are reported; frequency estimates and genotype–phenotype correlations are provisional. Many cohort-level statistics (imaging, seizure onset, mortality) derive from **cross-gene IGD cohorts**, not PIGB-specific data — an extrapolation caveat.
- 2. Neurodevelopmental branch is inferred.** The link from GPI-AP loss to cortical malformation and epilepsy rests on the established biology of individual GPI-APs (glypicans, contactins, RGMa) rather than direct demonstration in PIGB-deficient neurons.
- 3. No faithful whole-animal or iPSC-neuronal model** of DEE80 is established; the neurodevelopmental phenotype has not been experimentally reconstituted.

4. **Vitamin B6 mechanism is partly hypothetical.** The TNAP–PLP rationale is biochemically sound, but efficacy is partial and no participant achieved seizure freedom; the precise contribution of disturbed B6 metabolism to PIGB-related seizures is not fully proven.
 5. **No prevalence/incidence, QoL, or long-term natural-history data** specific to DEE80.
 6. **Epigenetic, metabolomic, and multi-omic** profiling of DEE80 tissues is essentially absent (2-oxoglutaric aciduria in severe cases is unexplained mechanistically).
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Proposed Follow-up Experiments / Actions

1. **Patient-derived iPSC cortical organoids / neurons** carrying PIGB variants to directly test whether reduced surface glypicans/contactins/RGMA cause migration and myelination defects — closing the inference gap in the neurodevelopmental branch.
 2. **PIGB conditional/hypomorphic mouse model** (neural-specific) to recapitulate cortical malformation, epilepsy and neuropathy, and to test pyridoxine/PLP and GPI-pathway-directed therapies in vivo.
 3. **Quantitative PLP/pyridoxal metabolomics** (serum and, where possible, CSF) in PIGB patients before/after high-dose B6, correlated with surface TNAP activity, to test the F007 mechanism directly and identify B6 responders.
 4. **Genotype–phenotype registry** aggregating all PIGB cases with standardized phenotyping, MRI (including central tegmental tract diffusion), ALP levels, and residual enzyme activity to build predictive prognostic models.
 5. **Systematic ClinVar/gnomAD curation** of PIGB variants with functional (CHO rescue / flow-cytometry) classification to improve ACMG interpretation and diagnostic yield.
 6. **Explore GPI-pathway or substrate-supplementation therapeutics** (e.g., strategies used in other CDGs) as candidate disease-modifying approaches, given the recessive single-enzyme etiology.
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Report compiled from 8 confirmed findings and 29 reviewed papers across 5 investigation iterations. Evidence types are labeled human clinical, model organism, in vitro, computational, or review throughout.

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